

Requirement Analysis

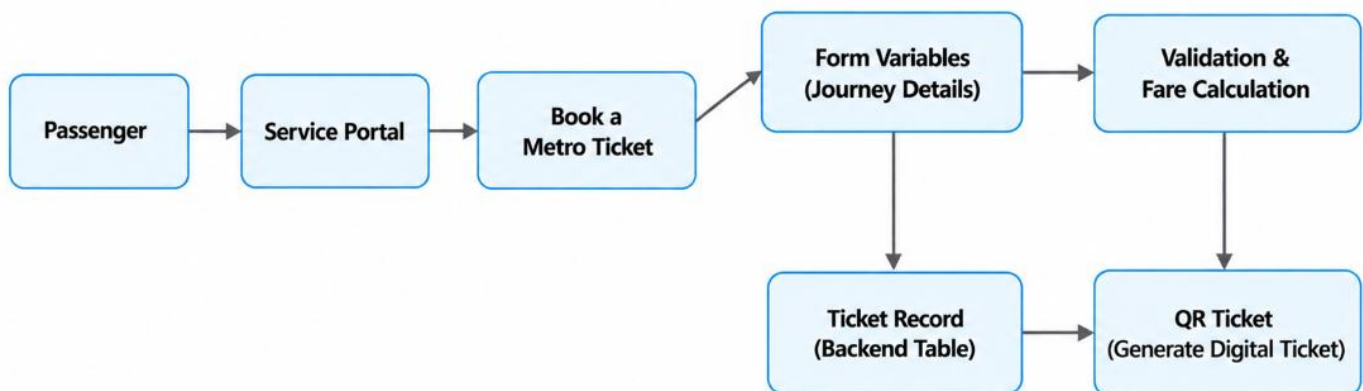
Data Flow Diagram & User Stories

Date	04 September 2026
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Project Name	Metro Ticket Generating System
Maximum Marks	

Logical Data Flow

Step	Flow
1	Passenger opens the ServiceNow Service Portal.
2	Passenger selects the Book a Metro Ticket service.
3	Portal displays the ticket-booking form with journey and passenger variables.
4	Passenger selects starting station, destination, journey type, passenger count, and payment mode.
5	Catalog Client Scripts and UI Policies validate inputs and calculate the applicable fare.
6	The ticket request is submitted and the booking information is stored in the backend table.
7	A digital QR ticket is generated for the submitted booking and can be used for verification.

Metro Ticket Generating System — Data Flow Diagram



Flow: Passenger → Service Portal → Book a Metro Ticket → Form Variables → Validation & Fare Calculation → Ticket Record → QR Ticket

Figure 1: High-level Metro Ticket Generating System data flow.

User Stories

User Type	User Story	Acceptance Criteria	Priority
Passenger	Open the Service Portal and find the metro ticket service.	Book a Metro Ticket item is visible and accessible.	High
Passenger	Select starting and destination stations.	Valid station values can be selected.	High
Passenger	Select the journey type and number of passengers.	Journey type and passenger count are captured correctly.	High
Passenger	View the fare based on the selected route.	Fare is automatically calculated according to the configured route and journey type.	High
Passenger	Enter payment information when required.	Payment mode field appears according to the selected option.	Medium
Passenger	Submit the metro ticket request.	Request is created successfully with the entered booking details.	High
Passenger	Receive a digital QR ticket.	QR code is displayed after ticket submission for verification.	High
Administrator	Maintain metro station and ticket configuration.	Station records and catalog configuration can be maintained.	Medium
Administrator	Control access to metro booking data.	Unauthorized users cannot directly access restricted backend data.	Medium